Light following robot without any microcontroller

The IoT Academy

Connecting The Unconnected

The IOT Academy

• **Experiment 1:-**

To design and construct a simple robot that follows a light source using one LDR, one relay module, and two DC motors — without the use of any microcontroller.

• **Material required :-**

Component	Quantity
LDR (Light Dependent Resistor)	1
DC Motors (Geared or Simple)	2
Relay Module (SPDT, 5V/6V)	1
NPN Transistor (e.g., BC547)	1
Resistor – 10kΩ	1
Resistor – 1kΩ	1
Flyback Diode (e.g., 1N4007)	1
Breadboard	1
Jumper Wires	As required
Castor Wheel	1
Robot Chassis or Base Plate	1
6V or 9V Battery + Clip	1
Tape, Glue, Screws	As required

Circuit Description:

1. The LDR and a 10kΩ resistor form a voltage divider.
2. The voltage from this divider controls the base of an NPN transistor (e.g., BC547).
3. When light intensity increases, the LDR's resistance decreases, increasing base voltage and switching on the transistor.
4. The transistor energizes the relay, which switches motor power between the left and right motor.
5. Based on light intensity, either the left or the right motor is powered, causing the robot to turn toward the light source.

Procedure:

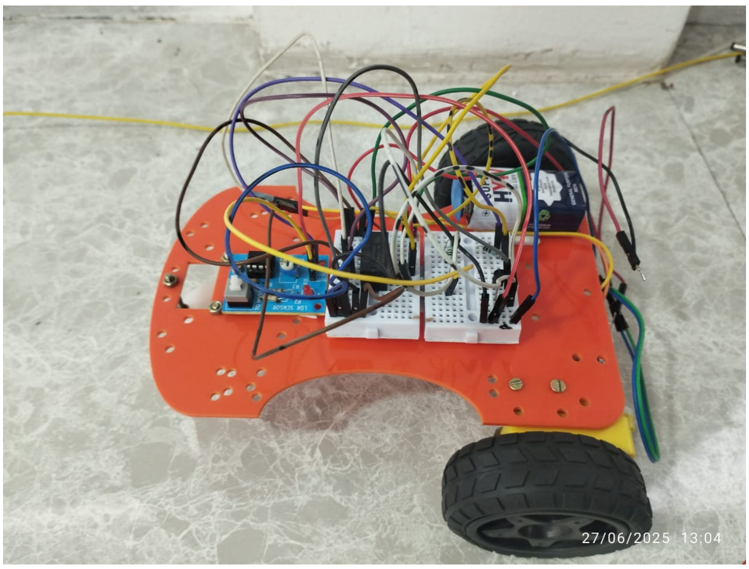
1. **Mounting:**
   - Attach the two DC motors to either side of the chassis.

    - Add the castor wheel for balance.
    - Secure the LDR facing forward (toward the light source).

2. **Building the Circuit:**
    - Connect the LDR in series with a 10kΩ resistor to form a voltage divider.
    - Connect the midpoint of the divider to the base of the NPN transistor through a 1kΩ resistor.
    - Connect the collector of the transistor to one side of the relay coil.
    - Connect the other side of the relay coil to +Vcc.
    - Place a flyback diode (1N4007) across the relay coil (cathode to Vcc).
    - Connect the emitter of the transistor to GND.
    - Connect the relay's COM pin to the battery positive.
    - Connect the NO and NC pins of the relay to the right and left motor terminals respectively.
    - Ground the other terminals of both motors.

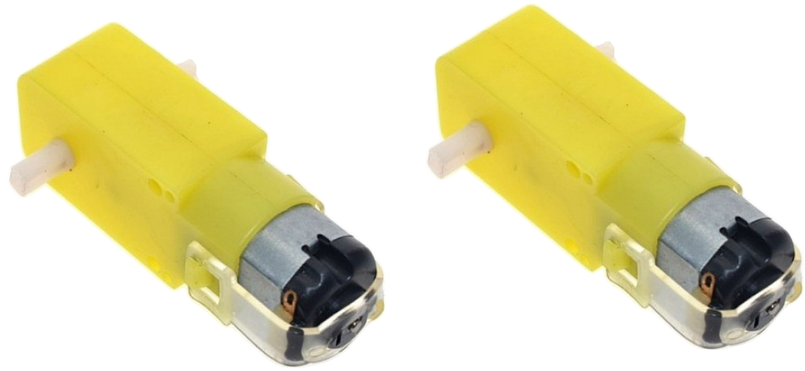
3. **Power On:**
    - Connect the battery (6V–9V) to the circuit.
    - Direct a flashlight or bright light source at the LDR and observe the robot movement.

PICTURE OF LIGHT FOLLOWING ROBOT :-



Component pic:-

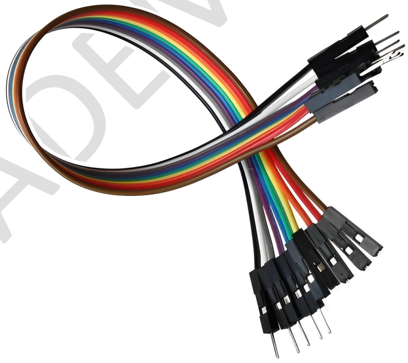
Dc motor



Relay

Ldc sensor

Jumper wire



Castor wheel

bread board

Battery

Pin Configuration & Connections

Component	Pin/Terminal	Connection
LDR	One end	+Vcc
	Other end	To 10kΩ resistor and 1kΩ (Voltage Divider)
10kΩ Resistor	One end	GND
	Junction with LDR	To base of transistor via 1kΩ resistor
1kΩ Resistor	One end	LDR + 10k junction
	Other end	Base of NPN transistor
NPN Transistor	**Base**	From 1kΩ resistor
	Collector	To one end of Relay Coil
	Emitter	GND
Relay Coil	One end	To Collector of NPN
	Other end	+Vcc
Flyback Diode	**Cathode (strip)**	To Vcc
	Anode	To Collector (Relay side)
Relay (Switch Side)	**COM (Common)**	Battery Positive Terminal
	NO (Normally Open)	Right Motor Terminal
	NC (Normally Closed)	Left Motor Terminal
Motors	Right Motor Other End	GND
	Left Motor Other End	GND

Observation Table:

Light on LDR	Relay State	Motor Running	Robot Direction
No light	OFF	Left motor	Turns right
Bright light	ON	Right motor	Turns left
Moderate light	Fluctuates	Fluctuates	Wiggles forward

Result:

The robot successfully followed the direction of a light source by switching between motors using a single relay, one LDR sensor, and basic components — without any microcontroller.

Conclusion:

This experiment demonstrates how basic electronics components such as LDRs, transistors, and relays can be used to mimic intelligent behavior in robots. The light following behavior was achieved by controlling motor direction based on the intensity of light falling on the sensor. No microcontroller or programming was required.